

Session 5: Regularization and Support Vector Machines (SVM)

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Objectives

- Understand the concepts of model capacity, underfitting and overfitting.
- Compare regularization mechanisms to deal with overfitting
- Analyse SVM as an extension of lineal models
- Understand the kernel trick

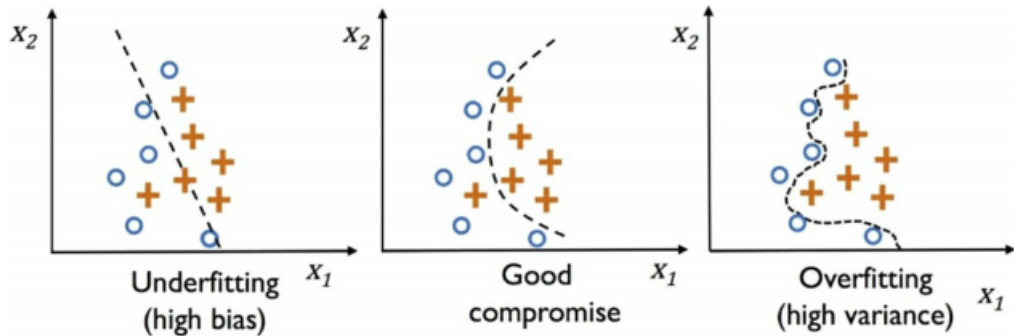
Summary

Model Capacity Overfitting and Underfitting

Regularization Methods

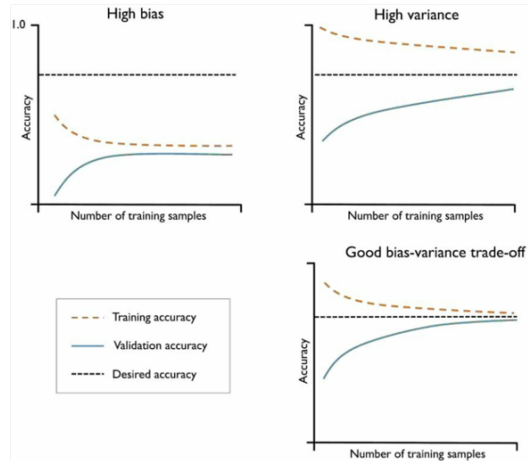
Support Vector Machine (SVM)

Problem: Overfitting & Underfitting



- A model can work well with training set but not with test set. **Do not generalize**

Learning curves



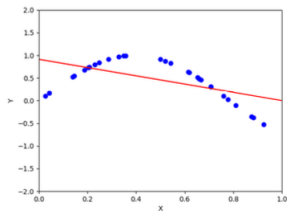
Regularization

Idea

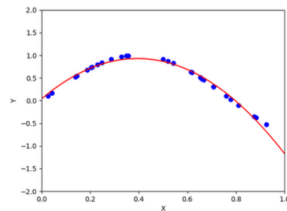
- Regularization limited the ability of a model to **reduce its specialization** or ability to memorize data.
- How can regularization further reduce the capacity of a linear model?
- A key point here is to keep in mind that a linear model does not necessarily have to make a linear fit to the data.
- For example, it is typical to make polynomial fits of order n .
- **Augmented dataset**

Polynomial Regression

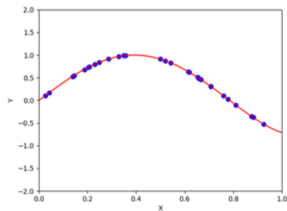
Fitting of the linear regression model with polynomial terms.



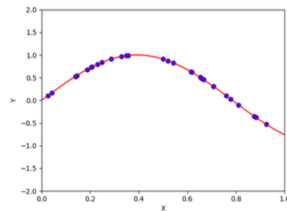
(a) Orden 1.



(b) Orden 2.



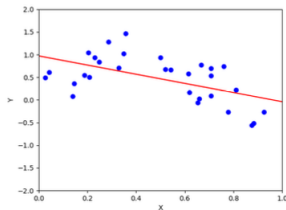
(c) Orden 4.



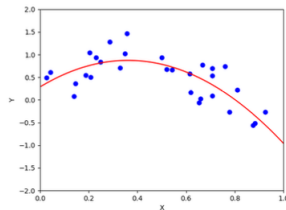
(d) Orden 20.

Polynomial Regression

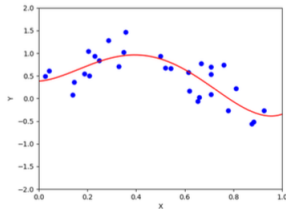
Fitting of the linear regression model with polynomial terms **with noise**



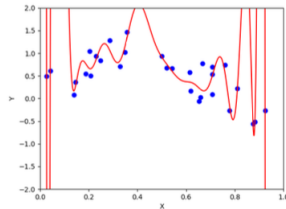
(a) Orden 1.



(b) Orden 2.



(c) Orden 4.

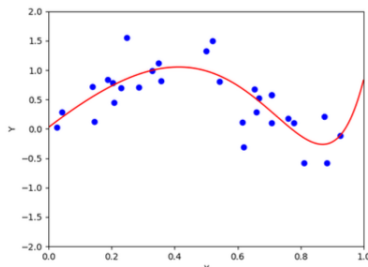


(d) Orden 20.

Regularization

How to proceed

- In practice we do not know how the data has been generated — > Difficult to adjust the model capacity
- Usually we will use complex models (danger of overfitting)
- ... and use regularization (example with order 20 and regularization)

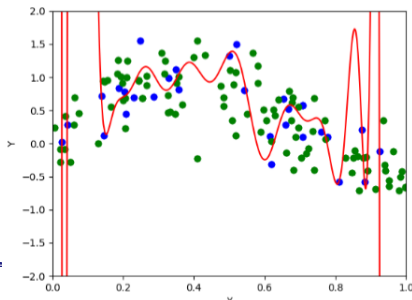


Regularization

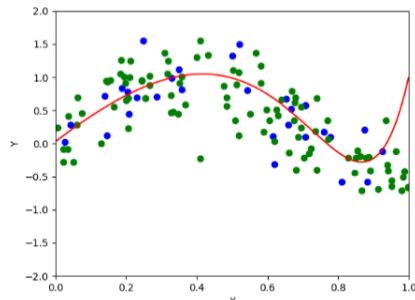
Training and test data on the polynomial linear regression model of order 20

- Plotting data is not usually possible (high dimension)
- At a numerical level within the algorithm we would realize that we are suffering from overfitting because there would be a disparity between the MSE obtained with the training data and the MSE obtained with the test data (higher in the second case).

Without regularization

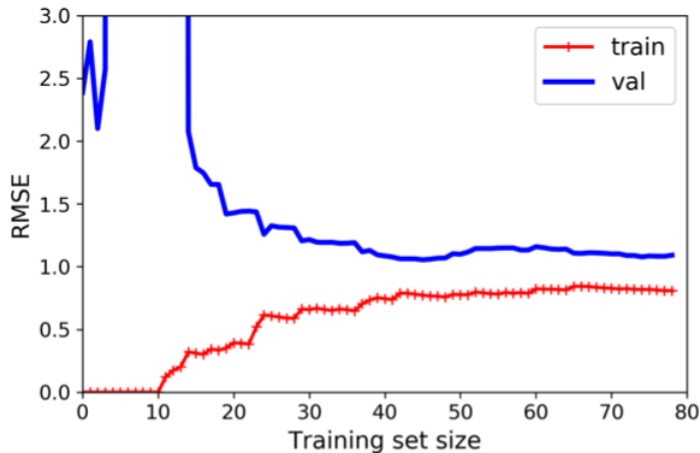


With regularization



Regularization

Learning Curves



Summary

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Regularization Methods

Support Vector Machine (SVM)

Regularization Methods

Regularization L1 or Least Absolute Shrinkage and Selection Operator (LASSO)

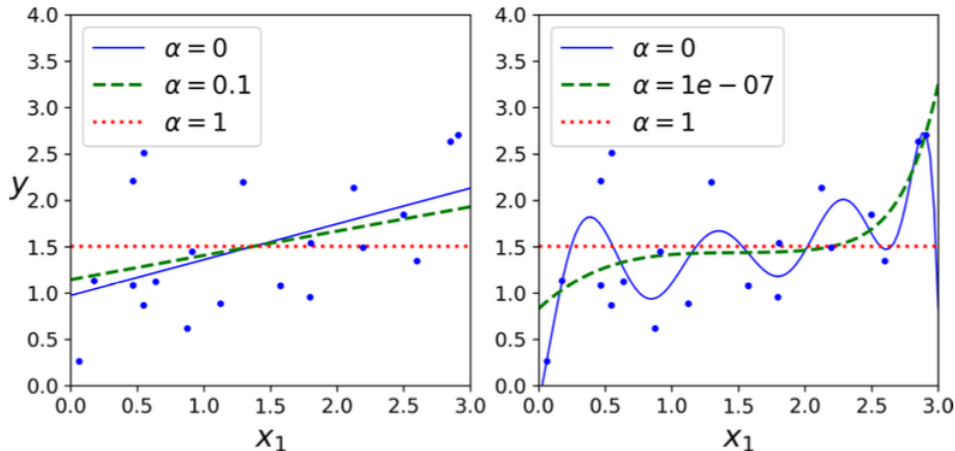
- We redefine the cost function that we optimize, adding to the MSE a term that penalizes the weights that we train with norm 1
- The cost function is

$$J(\boldsymbol{\theta}) = \text{MSE}(\boldsymbol{\theta}) + \alpha \sum_{i=1}^n |\theta_i|$$

- the algorithm will try to keep the weights at the smallest possible values
- An important characteristic of Lasso Regression is that it tends to eliminate the weights of the least important features (i.e., set them to zero)

Regularization Methods

A linear model (left) and a polynomial model (right), both using various levels of Lasso regularization



Regularization

Regularization L2 or Ridge

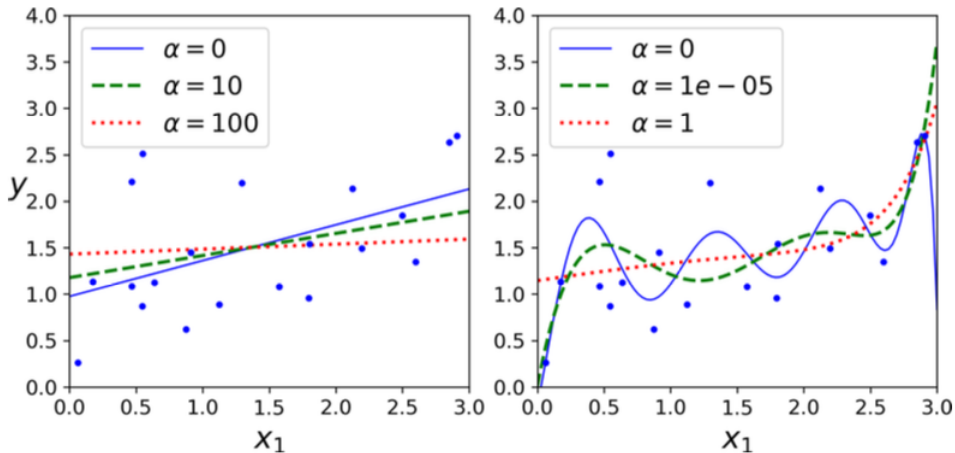
- Ridge Regression Cost function

$$J(\boldsymbol{\theta}) = \text{MSE}(\boldsymbol{\theta}) + \alpha \frac{1}{2} \sum_{i=1}^n \theta_i^2$$

- we can perform Ridge Regression either by computing a closed-form equation or by performing Gradient Descent

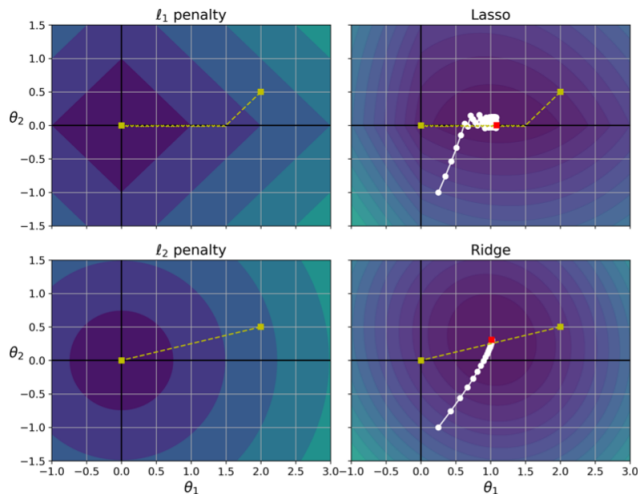
Regularization Methods

A linear model (left) and a polynomial model (right), both using various levels of Ridge regularization



Regularization Methods

Lasso vs. Ridge Regularization



Regularization

Combining L1 and L2: Elastic Net

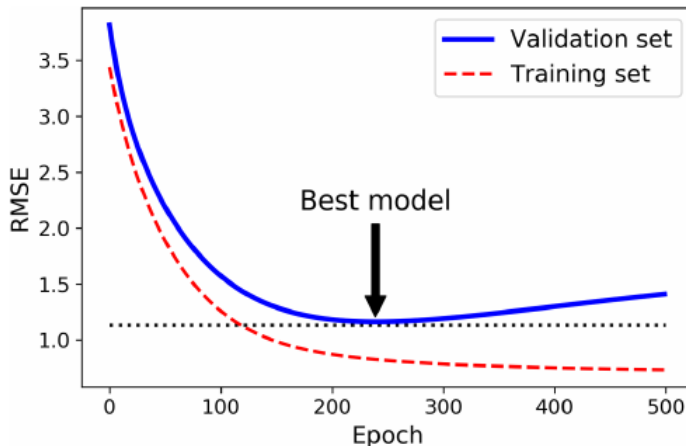
- Combining regularization L1 and L2 we get

$$J(\boldsymbol{\theta}) = \text{MSE}(\boldsymbol{\theta}) + r\alpha\sum_{i=1}^n |\theta_i| + \frac{1-r}{2}\alpha\sum_{i=1}^n \theta_i^2$$

- Usually it is preferable to have at least a little bit of regularization
- Ridge is a good default
- If you suspect that only a few features are useful, you should prefer Lasso or Elastic Net

Early Stopping

- Another way to regularize iterative learning algorithm (GD)
- Stop training as soon as the validation error reaches a minimum.



Summary

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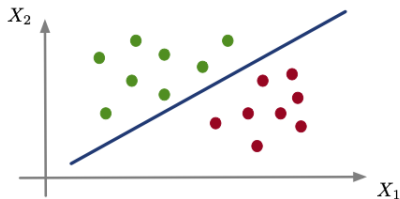
Support Vector Machine (SVM)

- SVM can be seen as one more step in linear models to achieve robust solutions, with a different approach to L1 or L2 regularization.
- There was a time when SVMs were even more promising than neural networks.
- It is a method that is still used since for some problems it offers us models with very good performance without the complexity of adjustment of neural networks
- Ee will focus on classification, what has undoubtedly generated much more literature.

Lineal SVM

Another approach to robust models

- Consider the following situation



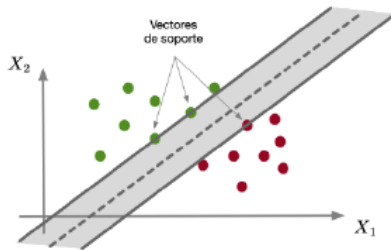
- The straight line is the result of applying logistic regression, so it is the result of optimizing the cross entropy

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(\hat{p}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{p}^{(i)}) \right]$$

Lineal SVM

Criterion for class splitting in SVM

- Is there other separation possibilities ?
- SVM try to plot a street as wide as possible



- The logic in this approach is to look for robustness in the separation since we look for the separation that gives us the most margin for the known points.

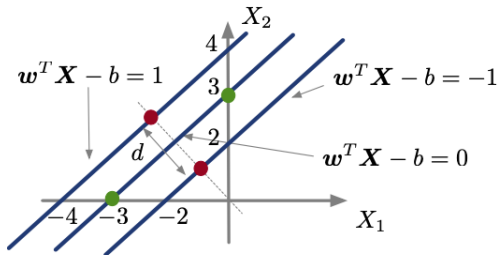
Lineal SVM

Minimization problem associate

- We have to find b and the vector w

$$\begin{aligned} & \min_{w,b} \quad ||w||^2 \\ & \text{sueto a} \quad w^T X^j - b \geq 1, \quad \forall j \text{ tal que } Y^j = 1, \\ & \quad \quad \quad w^T X^j - b \leq -1, \quad \forall j \text{ tal que } Y^j = -1 \end{aligned}$$

- Graphically



SVM

Soft Margin Classification

- The objective is to find a good balance between keeping the street as large as possible and limiting the *margin violations*
- The soft margin version is set as

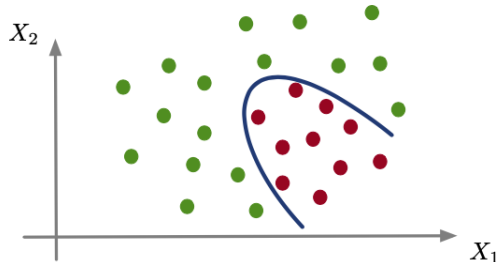
$$\min_{\mathbf{w}, b} \left\{ \left[\frac{1}{J} \sum_{j=1}^J \max(0, 1 - Y^j(\mathbf{w}^T \mathbf{X}^j - b)) \right] + \lambda \|\mathbf{w}\|^2 \right\}$$

- The value λ , which must be positive, will establish the importance of complying with the restrictions versus widening the street.

Non Lineal SVM

The kernel trick

- In a situation like the following it is impossible to separate both classes with a line.

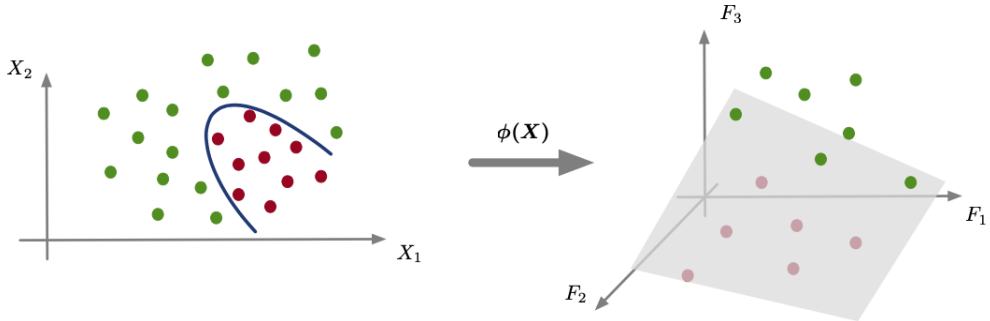


- Now we need the idea of the "kernel trick" to get SVM with non-linear regions

Non Lineal SVM

The kernel trick: The basic Idea

- The basic idea is to apply a non-linear transformation $\Phi(X)$ on the data that in turn increases the dimensionality, with the hope that in this new higher dimensional space, the data are perfectly separable by a hyper-plane.



Non Lineal SVM

The kernel trick: The Dual Problem

- The problem is the same than previous one but changing X^j by $\Phi(X^j)$
- This would be computationally hard
- But applying the Dual Problem from Optimization Theory we can express the problem in a convinient way.
- It is not necessary to make the data projections individually but it is enough for us to know the scalar product of all the transformed pairs, that is, $\Phi(X^j)^T \cdot \Phi(X^i)$

Non Lineal SVM

The kernel trick: Kernels

- The K kernels would then be functions of data pairs that satisfy this separability condition, that is, they would be functions that satisfy

$$K(X^j, X^i) = \phi(X^j)^T \cdot \phi(X^i)$$

- We give a couple of examples of the most used kernels:
 - **Gaussian Kernel** or Radial Basis Function (RBF): $K(X^j, X^i) = e^{-\gamma \|X^j - X^i\|^2}$
 - **Polynomial Kernel**: $K(X^j, X^i) = \phi((X^j)^T \cdot X^i)^d$
- In addition, they must meet another condition called Mercer's condition.
- The great advantage now is that to obtain the parameters ω^+ , b^+ of the hyperplane in the projection subspace it is not necessary to calculate the projection itself.

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